Formation 3 :

Evaluation du modèle



Plan :

- Introduction au ML
- Preprocessing
- Split en train/test
- Entraînement du modèle
- Evaluation et amélioration

Machine Learning Types in Simlytiks®

Supervised Learning

Regression

EXAMPLES:



Linear



Random Forest



Ridge

Classification

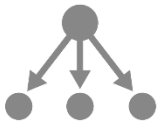
EXAMPLES:



Decision Tree



Feature Importance



Gaussian Naive Bayes

Unsupervised Learning

Clustering

EXAMPLES:



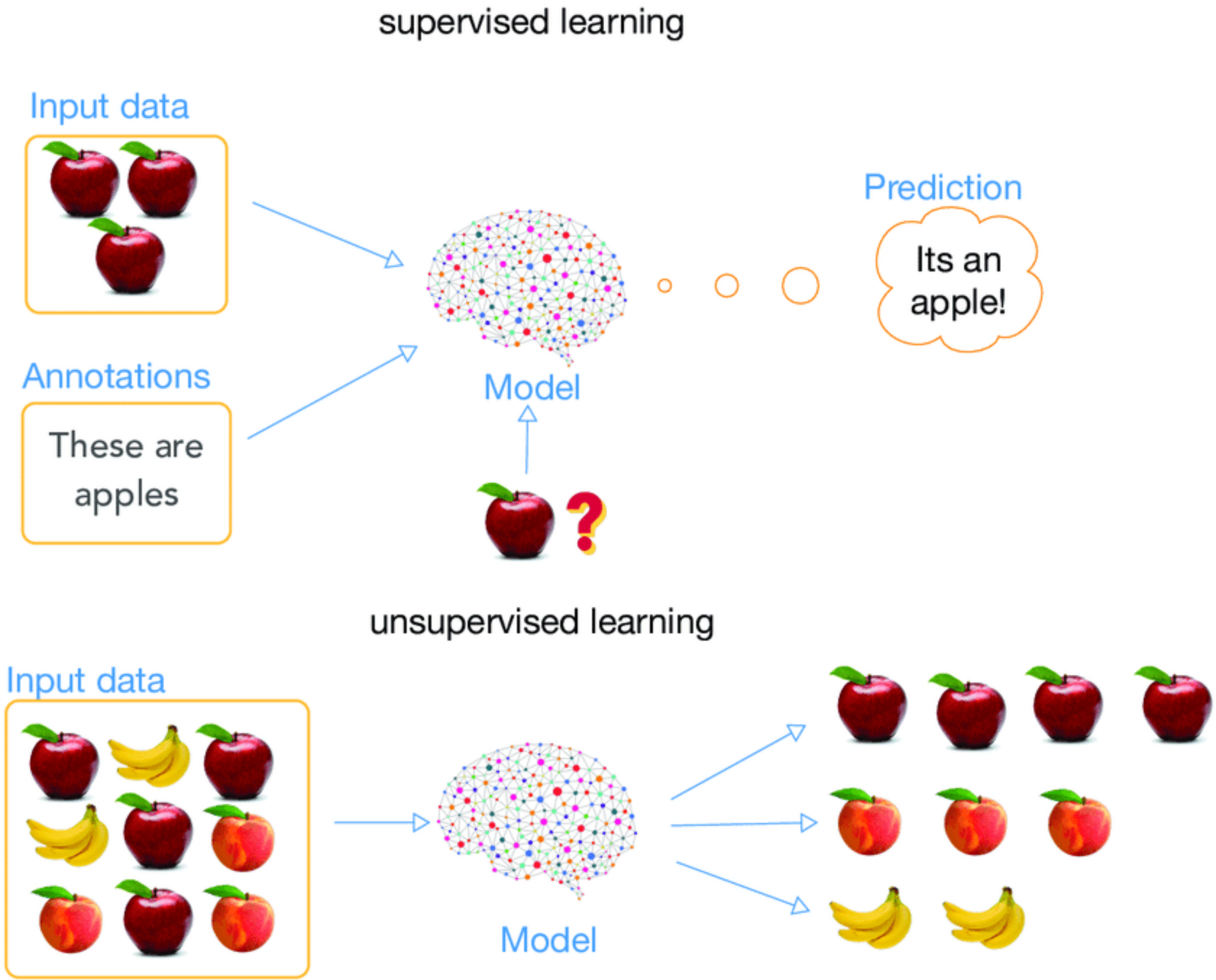
Mean Shift



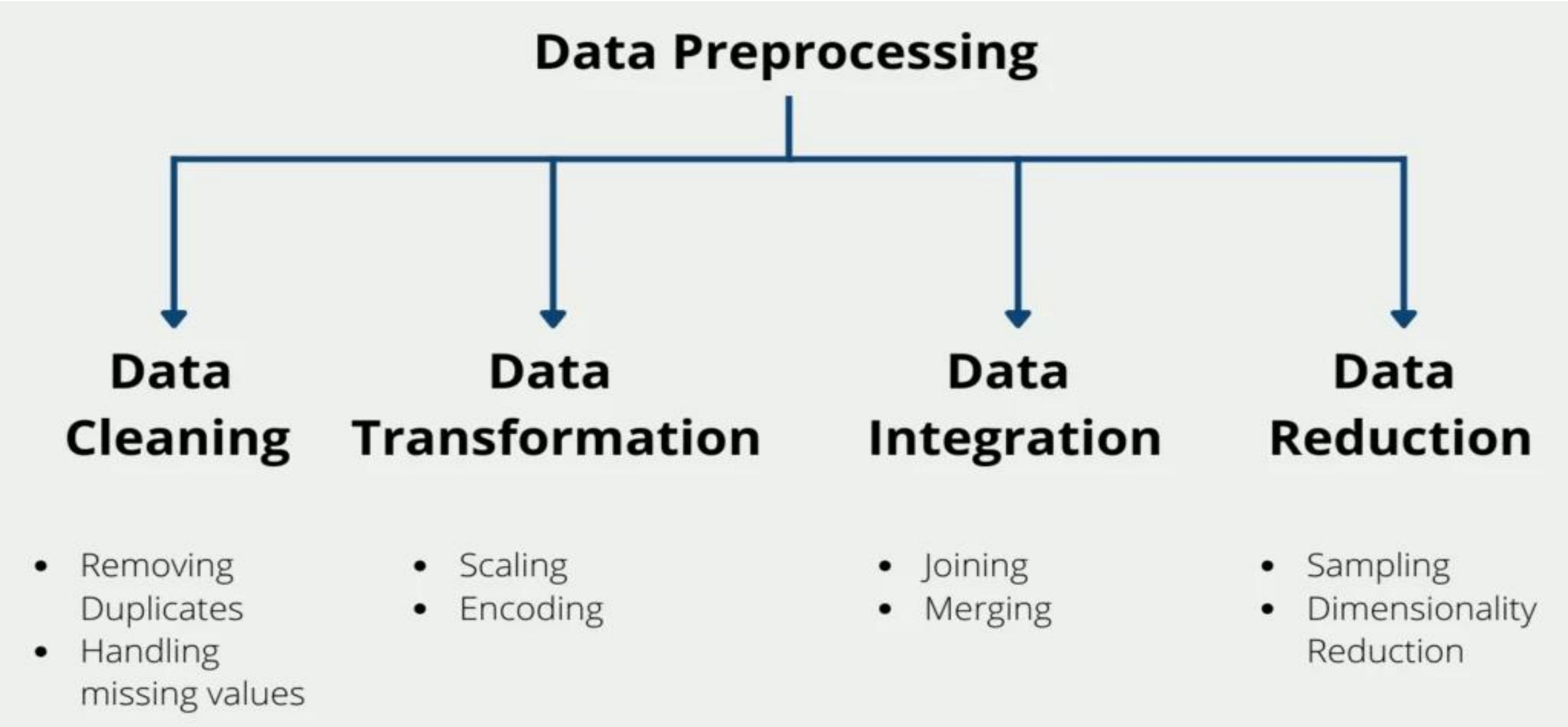
Louvian



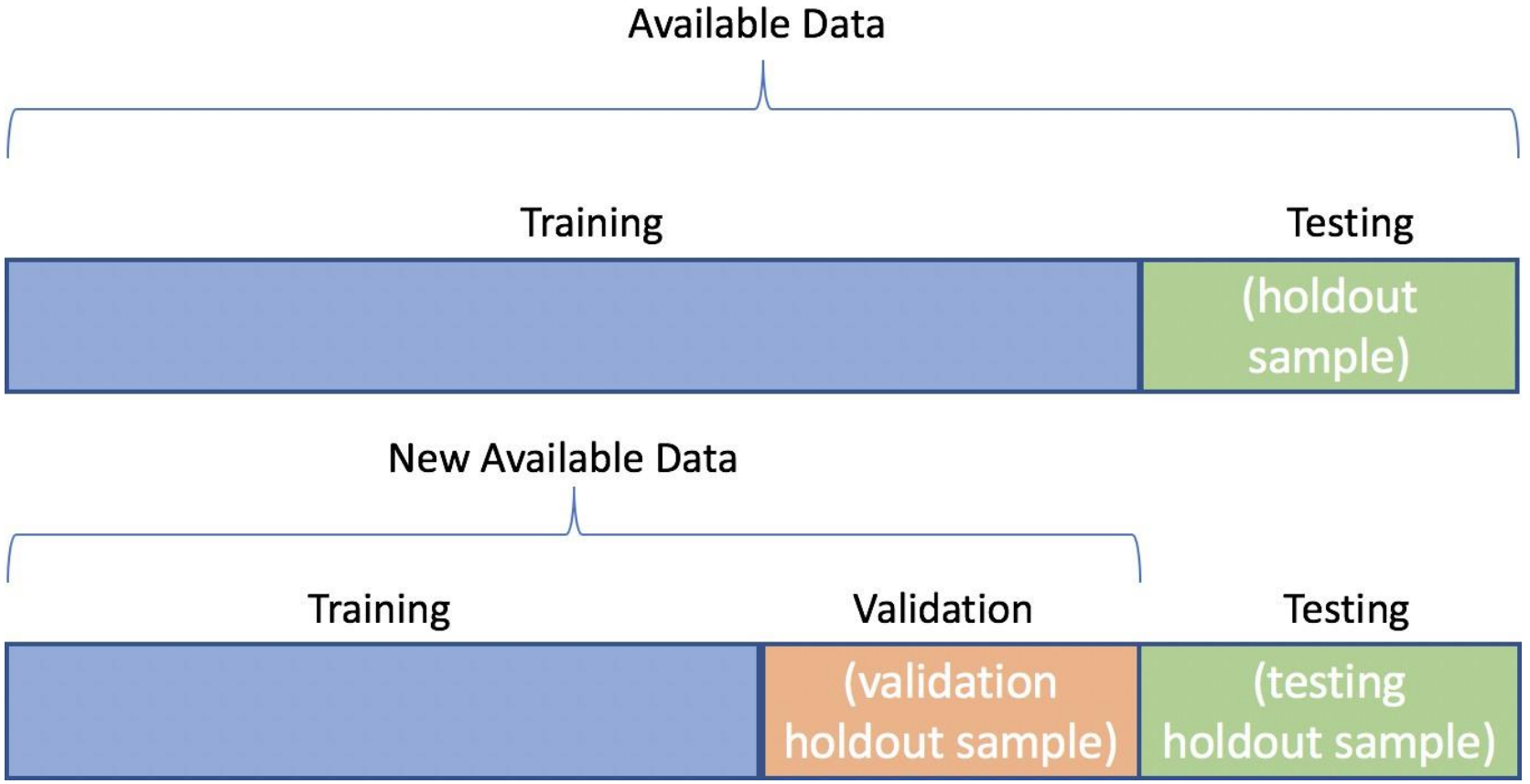
Hierarchical

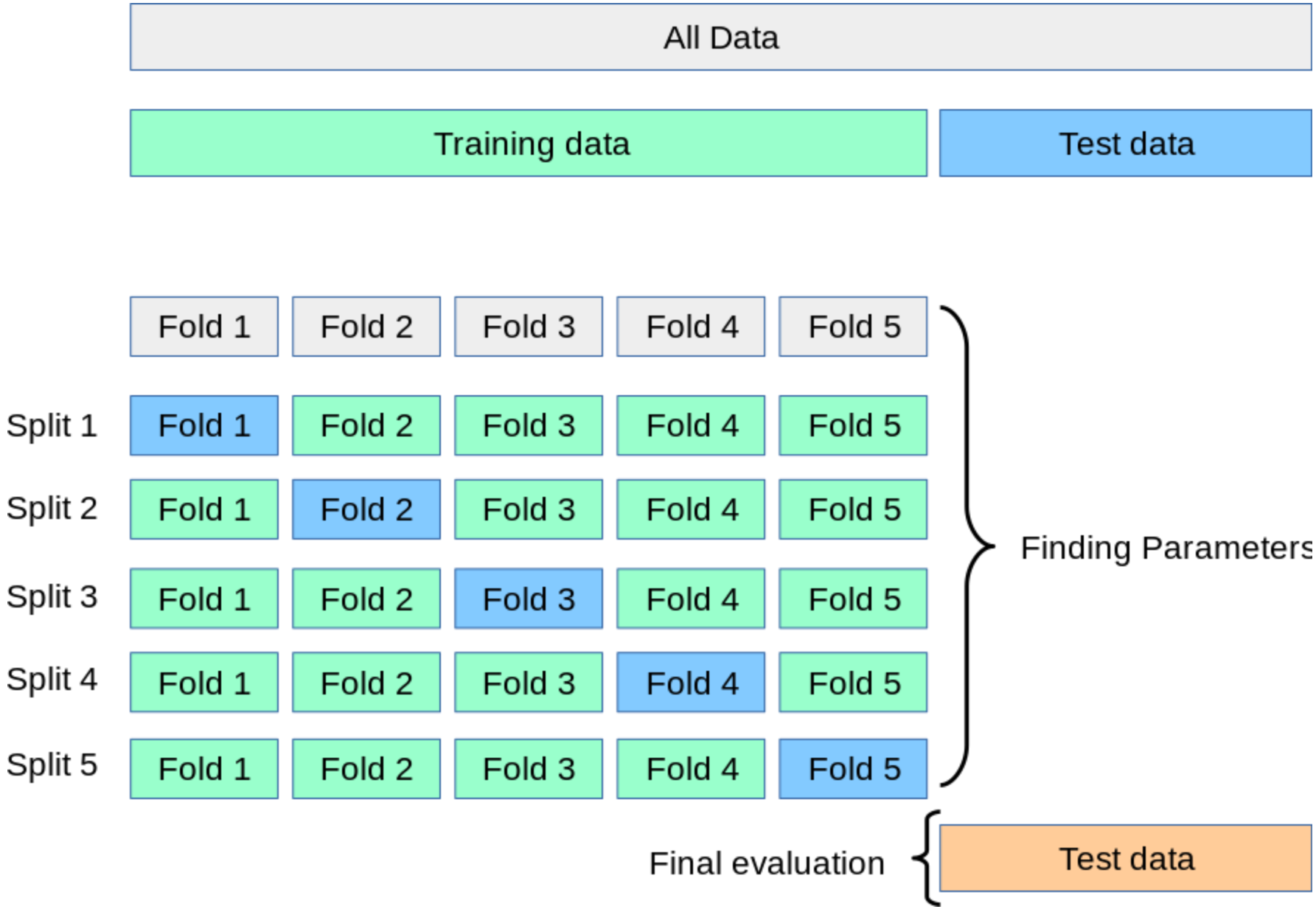

supervised learning
Input data
Annotations
These are apples
Model
Prediction
Its an apple!
unsupervised learning
Input data
Model

L’étape la plus importante du ML :



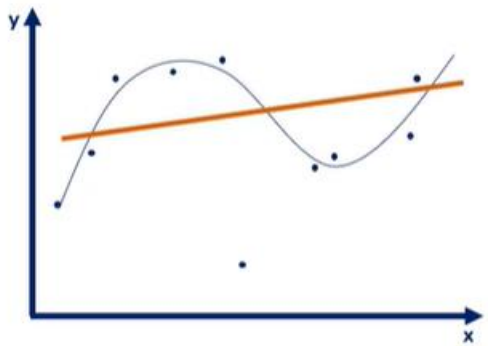
- Training set vs Test set




All Data
Training data
Test data
Fold 1
Fold 2
Fold 3
Fold 4
Fold 5
Split 1
Split 2
Split 3
Split 4
Split 5
Finding Parameters
Final evaluation
Test data

Underfitting and overfitting

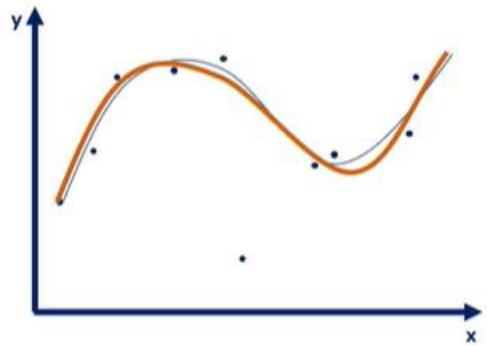
An **underfitted** model



Doesn't capture any logic

- High loss
- Low accuracy

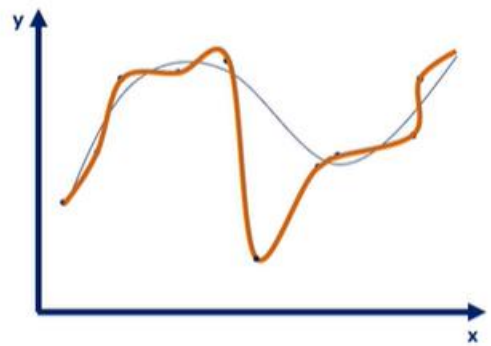
A **good** model



Captures the underlying logic of the dataset

- Low loss
- High accuracy

An **overfitted** model



Captures all the noise, thus "missed the point"

- Low loss
- Low accuracy

365 DataScience

Evaluer un modèle de régression :

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

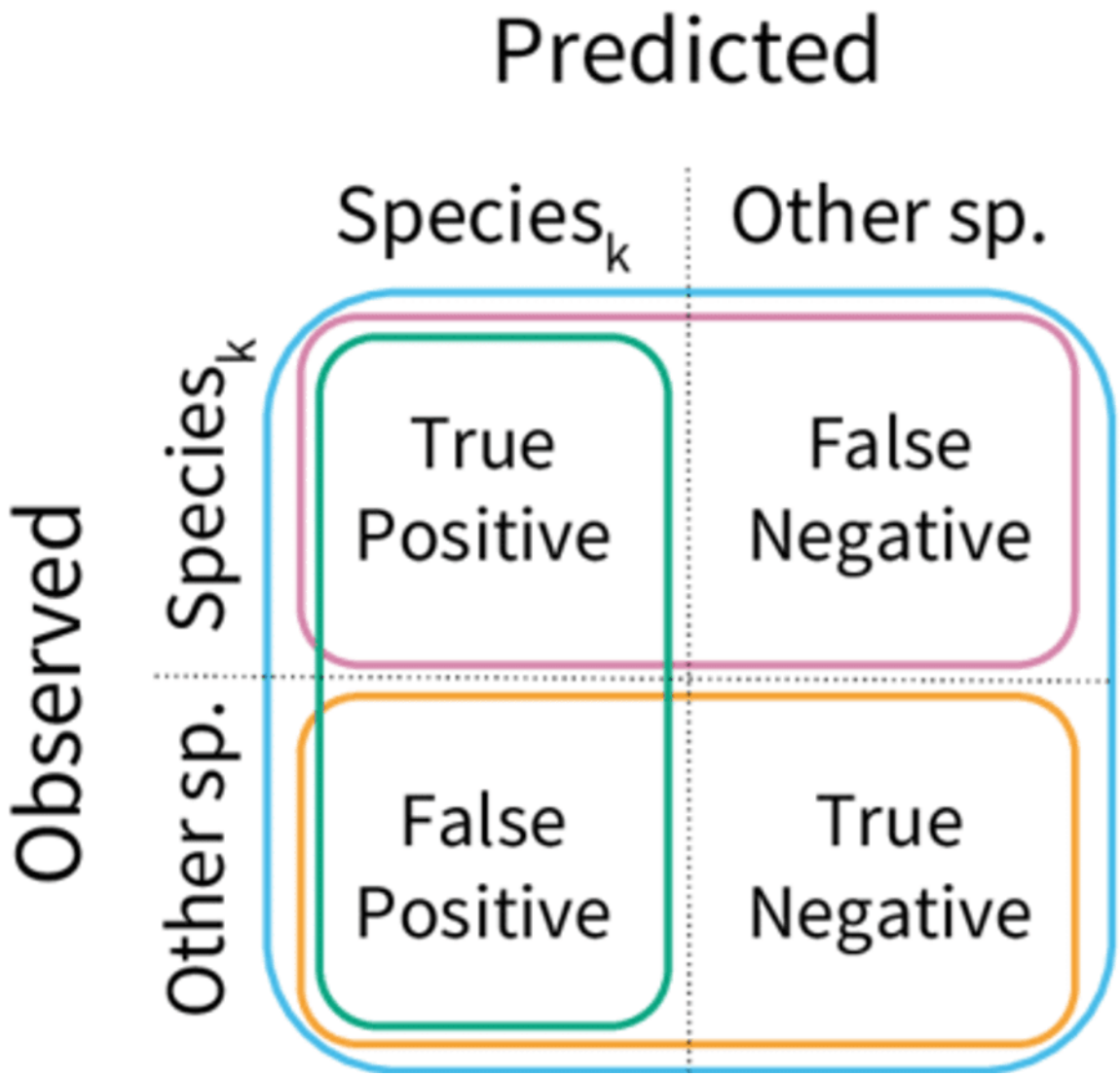
$$RMSE = \sqrt{MSE}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

Evaluer un modèle de classification : accuracy

$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{All predictions}}$$



Accuracy $= \frac{TP + TN}{TP + TN + FP + FN}$

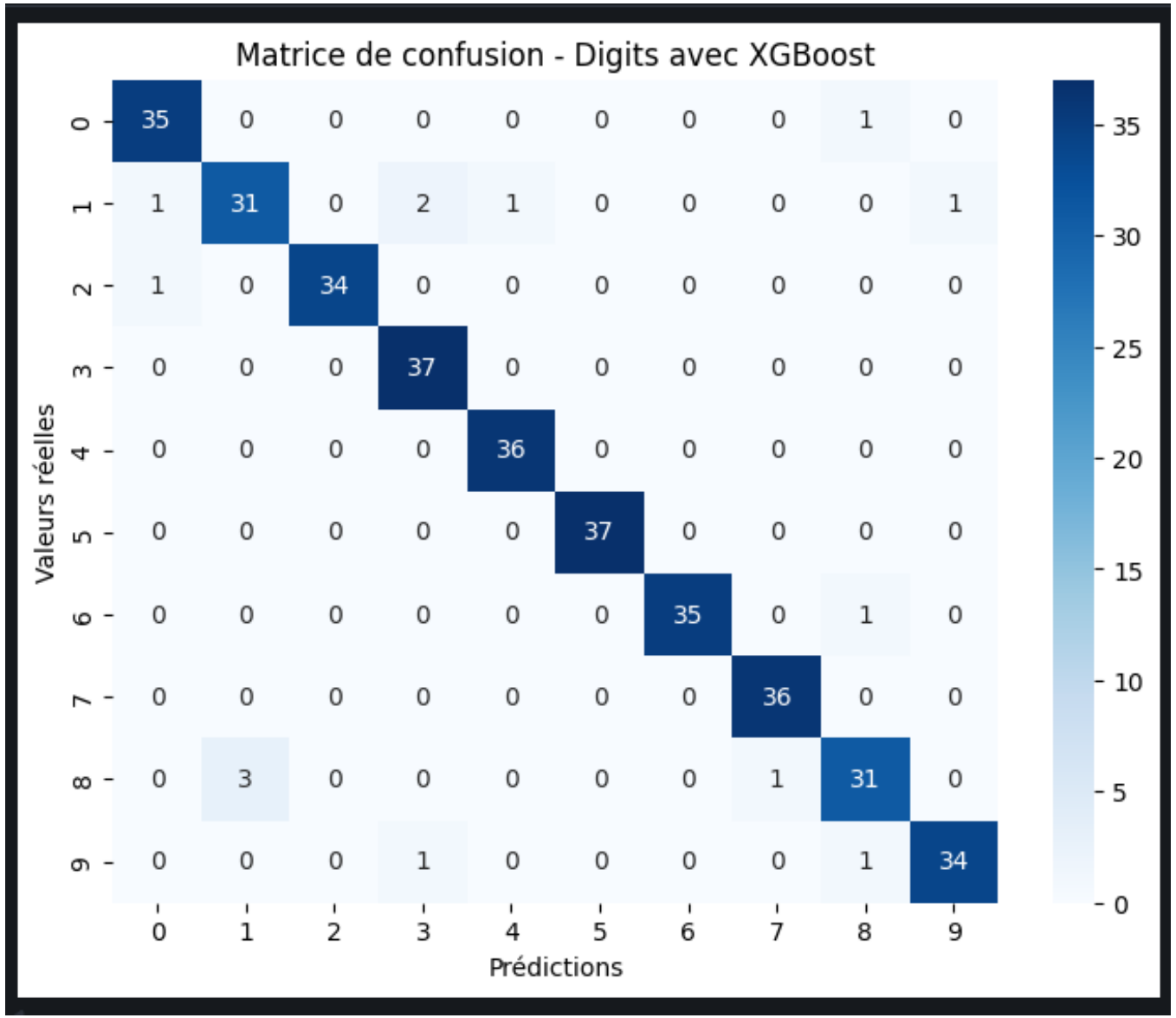
Specificity $= \frac{TN}{TN + FP}$

Precision $= \frac{TP}{TP + FP}$

Recall $= \frac{TP}{TP + FN}$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Confusion Matrix



A vous de jouer !